Worksheet 3-1 C The Chain Rule #3
AP Calculus Name_____

Directions: Find the derivative of the following functions. Simplify your answers!

1. $f(x) = -2e^{x^2-3x}$

2. $g(x) = 4e^{2-5x}$

3. $h(w) = we^{4w}$

4. $f(x) = \dfrac{e^{2x}}{5x}$

5. $y = e^{-x}$

6. $y = \sin\,(e^{3x})$

7. $f(x) = \ln|4x - 1|$

8. $g(x) = \ln|x^2|$

9. $h(x) = \ln|\sin\,(x)|$

10. $k(x) = \ln|e^x|$

11. $g(x) = \ln|3x^2 - 4x + 2|$

12. $m(x) = \ln|4 - x|$

13. $f(x) = e^{\cos(3x)}$

14. $h(x) = e^{e^{2x}}$

15. $g(x) = e^{\ln|x|}$

16. $y = \sin(\ln|3x + 2|)$

17. $f(r) = \left(\frac{r}{r-1}\right)^3$

18. $h(t) = \frac{1}{2}t - 2\sin^3(2t)$

19. $f(x) = \frac{1}{\sqrt[3]{x^2 - 1}}$

20. $F(t) = \sqrt{\frac{t}{t+1}}$

21. Find an equation of the tangent line to the curve $y = \sqrt{1 + x^3}$ at $(2, 3)$.